



13. Which of the following are not the characteristics of under reinforced section?  
 A) large deflections and wide cracks B) small deflections and narrow cracks  
 C) cracks develop at soffit and progress towards compression side D) considerable warning before failure
14. Effective reinforcement ratio in PSC design is based on:  
 A) Steel area and depth B) Concrete grade only  
 C) Anchorage zone stress D) Load ratios
15. Which of the following flexural flexures is generally not desirable?  
 A) under reinforced sections B) anchorage failure  
 C) fracture of steel in tension D) over reinforced sections
16. A web shear crack is more likely in:  
 A) Slabs B) T-beams  
 C) Thin web beams under heavy concentrated load D) Anchored piles
17. Increasing eccentricity in prestressing leads to:  
 A) Greater tensile stresses B) Lower flexural efficiency  
 C) Lower shear strength D) Higher bending resistance
18. A post-tensioned beam designed for large span is usually:  
 A) Symmetrical I-section B) T-section with eccentric tendon profile  
 C) Rectangular solid section D) Circular section
19. A tendon profile with parabolic eccentricity is adopted to:  
 A) Resist concentrated loads at mid-span B) Reduce creep losses  
 C) Resist uniform loads D) Prevent end anchor slip
20. The most efficient use of prestressing steel is obtained when:  
 A) Tendons are concentric B) Tendons are placed in straight line  
 C) Tendons are distributed uniformly D) Tendons are eccentric and follow bending moment diagram

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                                                                                                                                                                                                                       | Marks | BT* | CO* | PO* |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) Illustrate why prestressed concrete structures require the use of high-strength steel and high-strength concrete.                                                                                                                                                                                                                                                                                  | 8     | L*2 | 1   | 1   |
|          | b) A rectangular concrete beam 100mm wide by 250mm deep spanning over 8m is prestressed by a straight cable carrying an effective prestressing force of 250kN located at an eccentricity of 40mm. The beam supports live load of 1.2kN/m. Construct the resultant stress distribution diagram for the centre-span cross-section of the beam assuming the density of concrete as 24kN/m <sup>3</sup> . | 8     | L3  | 2   | 2   |
| 2.       | a) Explain with neat sketches the step-by-step mechanism involved in pre-tensioning and post-tensioning of a concrete beam.                                                                                                                                                                                                                                                                           | 8     | L2  | 1   | 1   |
|          | b) A post-tensioned rectangular beam with section 300mm × 600mm is stressed with an initial prestress of 900N/mm <sup>2</sup> . There are 4 straight cables of each 200mm <sup>2</sup> with eccentricity 200mm. Determine the percentage loss of prestress due to elastic deformation in concrete, when the cables are pulled one by one. Take modular ratio, $m = 6$ .                               | 8     | L3  | 2   | 2   |
| 3.       | a) Explain the load balancing concept and pressure line concept with neat sketches.                                                                                                                                                                                                                                                                                                                   | 8     | L2  | 1   | 1   |
|          | b) A concrete beam with a single overhang is simply supported at A and B over a span of 8 m and overhang portion BC is 2 m. The beam is 300 mm × 900 mm and supports a UDL of 3.52 kN/m. Determine the profile of the cable to balance dead load and live load. Take prestressing force of 250 kN and density of concrete as 24 kN/m <sup>3</sup> .                                                   | 8     | L3  | 2   | 2   |

## Unit – II

- |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |    |   |   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 4. a) | An 8m span concrete beam having a rectangular section 150mm × 300mm is pre-stressed by a parabolic cable at an eccentricity of 75mm at midspan and zero eccentricity at supports. The effective prestressing force is 350kN. The beam supports a concentrated load of 20kN at centre span in addition to its self-weight. Find the short-term deflection at centre of span under pre-stress, self-weight and imposed load. Also find the long-term deflection if the loss ratio is 0.8 and creep coefficient is 1.6; Take $E_c = 38\text{kN/mm}^2$ ; density of concrete = $24\text{kN/m}^3$ | 8 | L3 | 3 | 2 |
| b)    | Explain the shear failures in prestressed concrete members and summarize the techniques to enhance their shear resistance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 8 | L2 | 4 | 1 |
| 5. a) | Derive an expression for deflection: (i) due to self-weight of a simple supported PSC beam. (ii) due to prestress for a PSC beam with straight cable.                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8 | L2 | 3 | 1 |
| b)    | A post tensioned bridge girder is of box section of overall dimension 1200mm wide and 1800mm deep with wall thickness of 150mm. The tensile steel has an area of $4000\text{mm}^2$ and is located at a distance of 200mm from the bottom of the section. Estimate the ultimate flexural strength of the section, if $f_{ck} = 40\text{N/mm}^2$ and $f_{pu} = 1600\text{N/mm}^2$ .                                                                                                                                                                                                            | 8 | L3 | 4 | 2 |
| 6. a) | A simple supported rectangular concrete beam of span 6 m with cross-section 150 mm × 300 mm is pre-stressed by a symmetric parabolic cable, at 120 mm from bottom fibre at the supports and at 60 mm from bottom fibre at midspan. The initial pre-stressing force in the cable is 300 kN. Taking the unit weight of concrete as $25\text{ kN/m}^3$ and modulus of elasticity of concrete as $3.8 \times 10^4\text{ N/mm}^2$ , determine the maximum deflection of the beam at transfer and the central concentrated force required to eliminate the above deflection.                       | 8 | L3 | 3 | 2 |
| b)    | Explain the flexural failures in prestressed concrete members for under-reinforced and over-reinforced sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 8 | L2 | 4 | 1 |

## Unit – III

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|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) | Design a simply supported bridge deck slab with the following data: Span = 10 m, slab width = 1 m, thickness = 500 mm. Permissible concrete compressive stress at transfer, $f_{ct} = 16.5\text{ MPa}$ ; Class I member (no tensile stress at any stage); safe stress in steel = $950\text{ MPa}$ ; live load = $10\text{ kN/m}^2$ ; prestress loss = 18%. Determine the required prestressing force and eccentricity. | 16 | L3 | 5 | 2 |
| 8. a) | Design a simply supported post-tensioned prestressed concrete beam with a span of 18 m and width 300 mm to carry a uniformly distributed load of $20\text{ kN/m}$ . The concrete stresses should not exceed 16 MPa in compression and 1.2 MPa in tension at any stage of loading. Considering an average prestress loss of 17%, design the beam depth, prestressing force, and eccentricity.                           | 16 | L3 | 5 | 2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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